

Product Requirements Document (PRD)

BizSight — AI-Powered Business Intelligence Platform

Product Name	BizSight
Tagline	Turning Data into Decisions
Document Type	Hackathon Project PRD
Version	1.0
Live URL	bizzsight.vercel.app
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1. Executive Summary

BizSight is a business analytics platform that converts raw business data (CSV/Excel) into actionable financial and operational insights — automatically. It targets small businesses, e-commerce sellers, and online stores that lack the technical or analytical skills to turn spreadsheet data into meaningful intelligence. The platform automates data cleaning, field mapping, financial analysis, and visualization, and layers AI-generated insights on top so business owners can make decisions without needing a data analyst.

2. Problem Statement

Small businesses, e-commerce sellers, and online stores generate valuable transactional data every day — but it typically sits in raw, unstructured Excel or CSV exports. Turning that raw data into meaningful business insight requires:

- Data cleaning and structuring skills
- Financial and statistical analysis knowledge
- Tooling (BI platforms, dashboards) that is often expensive or complex

Most small business owners have none of the above, so the data goes unused, and decisions are made on gut feeling rather than evidence.

3. Goals & Objectives

Goal	Description
Automate ingestion	Accept messy, inconsistently-labeled CSV/Excel files without requiring a fixed template
Remove technical barrier	Automatically detect and map business fields instead of asking users to reformat their data
Deliver real financial metrics	Calculate the core KPIs a business owner actually needs to make decisions
Make insights human-readable	Convert numbers into plain-language recommendations via AI, not just charts
Keep a human in the loop	Let the user verify/correct automated mapping before analysis runs

4. Target Users

Persona	Needs
Small business owner	Wants to understand monthly performance without hiring an analyst
E-commerce seller (Shopify/WooCommerce-style exports)	Needs to reconcile orders, returns, and profit from raw order exports
Online seller / solopreneur	Wants a fast, free way to see revenue and profit trends

5. Scope

In Scope (Hackathon MVP)

- CSV/Excel upload
- Automated data cleaning (missing values, duplicates, inconsistent columns)
- LLM-assisted column mapping with fuzzy-matching fallback (RapidFuzz)
- Manual mapping review/correction step
- Core financial metric calculation
- Interactive dashboard with charts
- AI-generated plain-language insights
- Export of processed analysis (CSV/Excel/PDF)

Out of Scope (Future Work)

- User accounts / authentication and saved history across sessions
- Multi-user/team collaboration
- Real-time data sync with live store platforms (Shopify, WooCommerce APIs)
- Mobile app
- Multi-currency/multi-language support

6. Functional Requirements

6.1 Data Upload

- Users can upload business data in **CSV** or **Excel** format.
- Upload is handled through a Next.js API route to the Python backend.

6.2 Data Cleaning & Processing

- Backend uses **Pandas** to clean and transform data: handling missing values, duplicate records, irrelevant columns, and inconsistent structures.
- **OpenPyXL** used for Excel-specific parsing.

6.3 Automated Column Mapping

- An **LLM** analyzes uploaded column headers and maps them to BizSight’s required business fields (e.g., “Buyer,” “Client” → “Customer Name”).
- **RapidFuzz** provides fuzzy string matching as a fallback/complement to the LLM mapping.
- Mapping suggestions are surfaced through a **Mapping Review** UI component.

6.4 Human-in-the-Loop Verification

- User reviews suggested mappings and can manually correct any field before proceeding.
- Confirmed mapping is sent back to the backend to trigger analysis.

6.5 Financial & Statistical Analysis

Backend calculates the following core metrics using Pandas: - Revenue - Gross Profit - Net Profit - Total Orders - Average Order Value (AOV) - Customer Acquisition Cost (CAC) - Repeat Purchase Rate - Return / Cancellation Rate

6.6 Dashboard & Visualization

- React-based interactive dashboard displays all calculated metrics.
- Charts visualize trends (e.g., revenue vs. profit over time).

6.7 AI-Generated Insights

- LLM converts calculated results into plain-language business insights and recommendations (e.g., “Revenue growth is outpacing operating costs — consider scaling top-performing products”).

6.8 Export

- Users can export the processed analysis in **CSV, Excel, or PDF** format for offline reporting.

7. Non-Functional Requirements

Category	Requirement
Usability	No technical/data background required to use the platform
Performance	Analysis pipeline should return results within a reasonable time for typical SME-sized datasets
Reliability	Mapping errors must be recoverable via manual review, not silent failures
Deployment	Application must be publicly accessible (deployed on Vercel)
Data handling	Uploaded files are processed temporarily; processed data is held in memory/temporary storage, not persisted long-term (MVP scope)

8. Technical Architecture

Layer	Technology
Frontend	Next.js, React, Tailwind CSS
Backend	Python, FastAPI
Data processing	Pandas, OpenPyXL
Intelligent mapping	LLM + RapidFuzz (fuzzy matching)
Deployment	Vercel

High-level flow: User → Next.js/React frontend → API layer → FastAPI backend → Pandas cleaning & analysis (+ LLM/RapidFuzz mapping) → Dashboard & AI insights → Export

9. User Flow (End-to-End)

1. **Upload Data** — User uploads a CSV/Excel file.
2. **Clean & Process** — Pandas cleans, transforms, and prepares the data.
3. **Auto Map** — LLM + RapidFuzz identify and suggest column mappings.
4. **Verify Mapping** — User reviews and confirms the mapping.
5. **Analyze** — Backend calculates key business metrics.
6. **Visualize** — Dashboard displays interactive charts and KPIs.
7. **Generate Insights** — LLM converts data into business insights.
8. **Export** — User downloads the analysis (CSV/Excel/PDF).

10. Success Metrics (Hackathon Demo Criteria)

Metric	Target
End-to-end demo	Upload → Insights → Export completes without manual intervention beyond mapping verification
Mapping accuracy	Correctly auto-maps common field-name variants (e.g., “Buyer”/“Client” → “Customer Name”)
Metric correctness	All 8 core financial metrics calculate correctly on sample dataset
Judge clarity	Dashboard and insights are understandable to a non-technical judge without narration

11. Team & Roles

Team Member	Role & Contribution
Manahil Asif	Team Lead — Prepared the presentation slides and visualizations, coordinated the project
Abdul Muqeet	Lead Developer — Built and deployed the complete application (frontend, backend, data processing, analytics)
Areeba Tahir	Live Demo & QA — Tested the application and presented the live demo
Faiza Batool	Research & Documentation — Originated the project idea and supported content documentation

12. Assumptions & Risks

Assumption / Risk	Notes
Users upload reasonably structured tabular data	Extremely malformed files may still require manual cleanup
LLM mapping requires internet/API access	Dependent on LLM API availability during the demo
Temporary storage is acceptable for MVP	No persistence layer means results don’t survive a page refresh (documented as future scope)

13. Future Scope

- User authentication and saved report history
- Direct API integration with e-commerce platforms (Shopify, WooCommerce)
- Scheduled/recurring analysis (e.g., weekly auto-reports)
- Multi-currency and multi-language support
- Team/collaborative workspaces